

Angela Qian

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EDUCATION

Purdue University | *B.S. in Computer Science, Minor in Mathematics*

Aug 2023 – May 2027

EXPERIENCE

Undergraduate Research Assistant | *Advisor: Joseph Campbell*

Feb 2025 – Present

- Currently leading a project developing hierarchical world models in model-based reinforcement learning
- Worked on a team to design an LLM-based agent with Theory of Mind to play the social deduction game Avalon
- Conducted participant studies with 45+ volunteers and applied statistical analysis to evaluate model performance
- Developed and evaluated methods for hallucination detection with 95% agreement with human annotations
- Drafted and co-authored a paper in submission to a top-tier machine learning conference

Software Engineering Intern | *Cepton Technologies Inc.*

May 2025 – Aug 2025

- Performed real-time foreground-background segmentation masking from depth and reflectivity maps by engineering a computer vision model based on U-Net segmentation, which has been integrated into the production system
- Built in Rust a real-time object tracking pipeline using LiDAR-generated point cloud data on a per-frame basis
- Developed a 2D rectangle selection tool for a 3D point cloud viewer, enabling users to select points from any camera perspective and display their coordinates dynamically by transforming 3D point data using projection matrices

Teaching Assistant | *Purdue University Department of Computer Science*

Jan 2025 – Present

- CS 381: Analysis of Algorithms, CS 211/311: Competitive Programming, CS 182: Discrete Mathematics
- Led weekly recitation sessions, teaching problem-solving techniques and reinforcing theory taught in class
- Held weekly office hours to clarify course material and guide students through homework challenges

Undergraduate Researcher | *Purdue University Vertically Integrated Projects*

Aug 2024 – Dec 2024

- Detected image blur with custom program using convolution, the Laplacian filter, and the Sobel filter
- Generated clear, focused photos from blurred images with algorithms based on Fourier Transforms
- Presented in Purdue's 2024 Fall Undergraduate Research Exposition

Competition Mathematics Instructor | *AlphaStar Academy*

July 2024 – Aug 2024

- Delivered lectures to 50+ students on Algebra, Number Theory, Geometry, and Counting/Probability
- Held review sessions with practice problems to prepare students for the AMC10, AMC12, and AIME exams
- Presented feedback to students and parents, highlighting focus areas for continued academic growth

PUBLICATIONS

[Bayesian Social Deduction with Graph-Informed Language Models](#)

S. Rahimirad, G. Gergerli, L. Romero, **A. Qian**, M. L. Olson, S. Stepputtis, J. Campbell. *NeurIPS Workshop on Efficient Reasoning* (2025), *Full version Under Review*

PROJECTS

[2048 AI Player](#) | *PyTorch, Jupyter Notebook, JavaScript*

June 2025 - Present

- Developed a neural network to play 2048 using imitation learning trained on a custom dataset of my own gameplay
- Built model in PyTorch to predict actions from board states, and exported it via ONNX to deploy in TensorFlow.js
- Authored and published a research-paper-styled [writeup](#) in SIGBOVIK-inspired journal, [SIGHORSE](#)
- Currently extending the project with reinforcement learning to fine-tune decision making through self-play

[ScribbleScore Handwriting Improvement](#) | *Python, OpenCV, Numpy, React, Flask*

Feb 2025

- Worked in a group of 4 over the course of a 36-hour hackathon (Boilermake XII)
- Designed bounding-box logic to group words and measure spacing consistency across clusters
- Implemented Sobel edge detection, line of best fit analysis to assess writing straightness, extracting text from images, and computing score

SKILLS

Languages: English (Native), Mandarin Chinese (Native)

Programming: Python, Rust, C/C++, JavaScript, Typescript, x86_64 Assembly, SQL, MongoDB, HTML, CSS, Java

Developer Tools: Git, GitHub, Jupyter Notebook, Figma, Linux, Overleaf, React, Node, Bash, LaTeX, VIM, FFmpeg, MySQL, MongoDB, Neo4J, Next.js, TailwindCSS, VS Code, Eclipse, Inkscape

Libraries: Matplotlib, OpenCV, PyTorch, TorchVision, Pandas, NumPy